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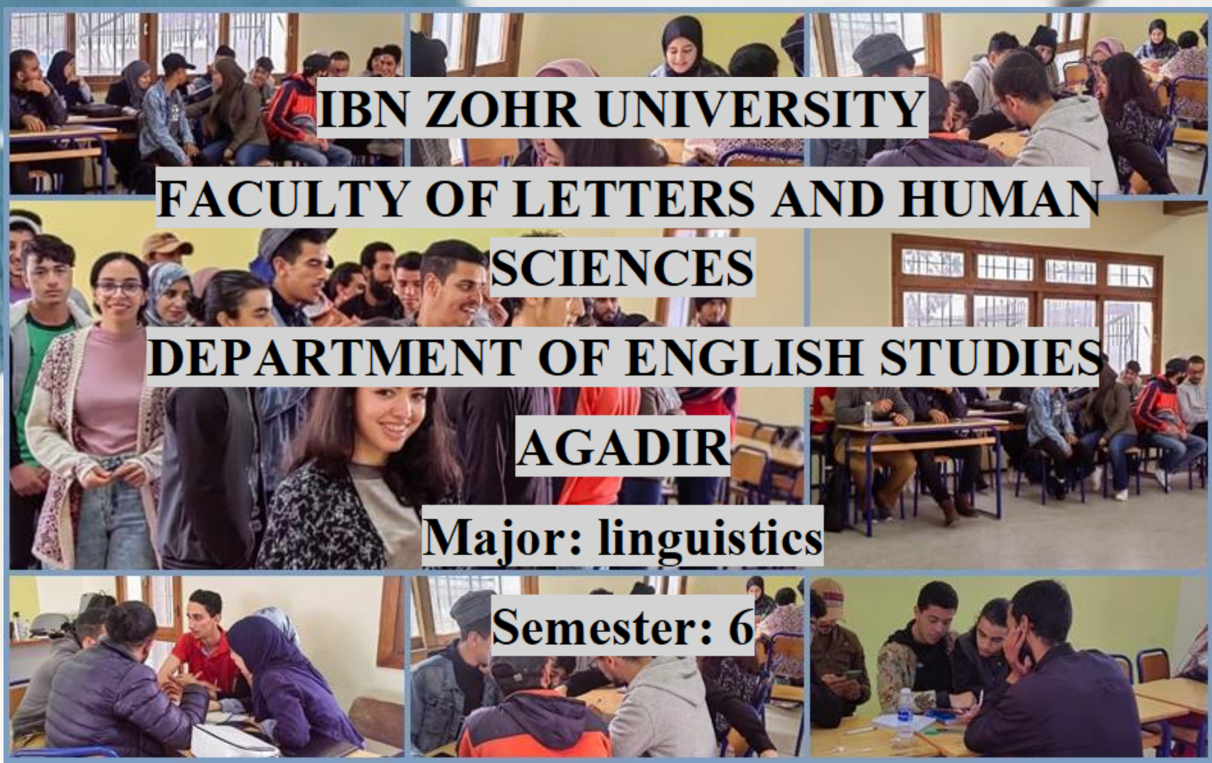

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SEMANTIC MAPPING AND SPACED REPETITION TO ACQUIRE NEW ENGLISH VOCABULARY IN AN AUTODIDACTIC COOPERATIVE GROUP.

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Table of Contents

SEMANTIC MAPPING AND SPACED REPETITION TO ACQUIRE NEW ENGLISH VOCABULARY IN AN AUTODIDACTIC COOPERATIVE GROUP.

ABSTRACT

The purpose of this study is to explore the impact of using Semantic Mapping and Spaced Repetition as a strategy to acquire new English vocabulary in an autodidactic cooperative group and to investigate the relationship between the vocabulary acquisition strategy preferred and their score on the vocabulary test. 16 students of the Department of English Language Studies of Ibn Zohr University, Agadir, Morocco took part in the study. Quantitative and qualitative data were collected through a two-part questionnaire The experimental group used semantic mapping and Anki as a memorizing techniques that is based on spaced repetition whereas the control group applied traditional techniques. The statistical results show that most students preferred note-taking as a strategy to acquire new vocabulary but the few who already preferred space repetition achieved the best scores in the Oxford English vocabulary level test. As the findings indicate, Semantic mapping and Space repetition seems to be an effective tool for vocabulary acquisition or in learning in general.

I. INTRODUCTION

Learning new vocabularies in the targeted language is very important for Language Acquisition. Teachers of L2 have concluded that knowing a language is widely related to knowing its vocabulary well. However, in the past vocabulary was given less importance. A vast part of communication depends on understanding words and remembering them. Therefore, we may say that learning a sconed language is a matter of knowing its vocabulary and being able to recall them at the needed moment.

Proficiency in a language is connected to how many words you can use. Without comprehending words, a Speaker or Hearer can't use the language effectively. There are many aspects to be considered for us to say someone knows a language. ''Knowing only the semantic meaning is a shortcoming." Dr. CHOUAIBI. A (2022). But still, this doesn't mean that words are not a crucial part of proficiency.

Language learning strategies (LLS) research began in the 1970s with Rubin and Naiman et al. (1978) attempting to find the characteristics of successful language learners to give insights for effective vocabulary teaching and learning. ''Research results revealed that no single method, instruction, or material could guarantee effectiveness on its own in foreign language learning". Dileka. Y, Yuruk. (2012). on-account-of, scholars, and researchers observed that many learners could not succeed whereas others could learn no matter what teaching method was used, studying the same material, in the same classroom, under the same conditions. Each person has their way of learning ''the so-called profile of intelligences -and in the ways in which such intelligences are invoked and combined to carry out different tasks, solve diverse problems, and progress in various domains." Gardner (1991). These challenges have opened up interests in the field of learning strategies.

The Focus on vocabulary in language learning and teaching has grown in n the last decade even though it has been placed as less priority in language teaching than the other skills, turns out vocabulary plays an important role in knowing a language. We can't ignore the fact that vocabulary is a significant element in a foreign language learning context. Taking the importance of vocabulary into consideration we cannot neglect the value of learning strategies in vocabulary learning and teaching. The British linguist. David Wilkins eloquently put it, "without grammar, very little can be conveyed; without vocabulary, nothing can be conveyed." The improved vocabulary offers valuable data and strengthen language abilities and may maximize language acquisition.

This research aims to know how to use the Semantic mapping strategy with Spaced Repetition to encourage self-sufficiency. Being aware of how the brain may store information would help us use these strategies effectively. For that reason, establishing strategies and expanding awareness of these vocabulary acquisition skills would be beneficial. It will assist students in overcoming their areas of weakness and be prepared for learning and give them the possibility of achieving their full potential. It will encourage flexible thinking, and be more analytic. Vocabulary learning strategies are therefore considered a means of empowering students to take control and responsibility for their learning.

Vocabulary learning strategies can be really helpful in learning and remembering a foreign language. By implementing the Semantic Mapping technique and then applying the Spaced Repetition for different vocabulary items, students can increase their vocabulary learning. This study will look into the learning strategies university of ibn zoher of the department of English studies prefers and their score on the vocabulary test to assess the better strategy. The main object of this experimental study is to examine whether using the Semantic Mapping Technique combined afterward with Spaced Repetition Technique is more efficient in helping retention of vocabulary in long-term memory than the Traditional Technique.

II. LITERATURE REVIEW

1. SEMANTIC MAPPING

This section is composed of two parts: theoretical and practical. Theoretically, I will try to explain the concept of semantic mapping and provide some examples. On the practical level, I will attempt to summarize the findings of relevant research studies and explain the procedures of creating a Semantic Map.

a. Theoretical Section

Semantic Maps (or graphic organizers) is a process of creating visual representations of categories and their relationships. The objective of creating a map is to graphically depict the meaning-based connections between a word or phrase and a group of related words or concepts. It can assist struggling students and individuals with disabilities in identifying, comprehending, and remembering the meaning of words they read in a text. It is a method of getting learners to connect new words to their own experiences and prior knowledge. “Semantic mapping is a visual strategy for vocabulary expansion and extension of knowledge by displaying in categories words related to one another" Kholi, & Sharifafar. An example of a semantic map can be found in figure.1. The structure of a semantic map in the example below includes a key concept with categorized concepts related to the main idea.

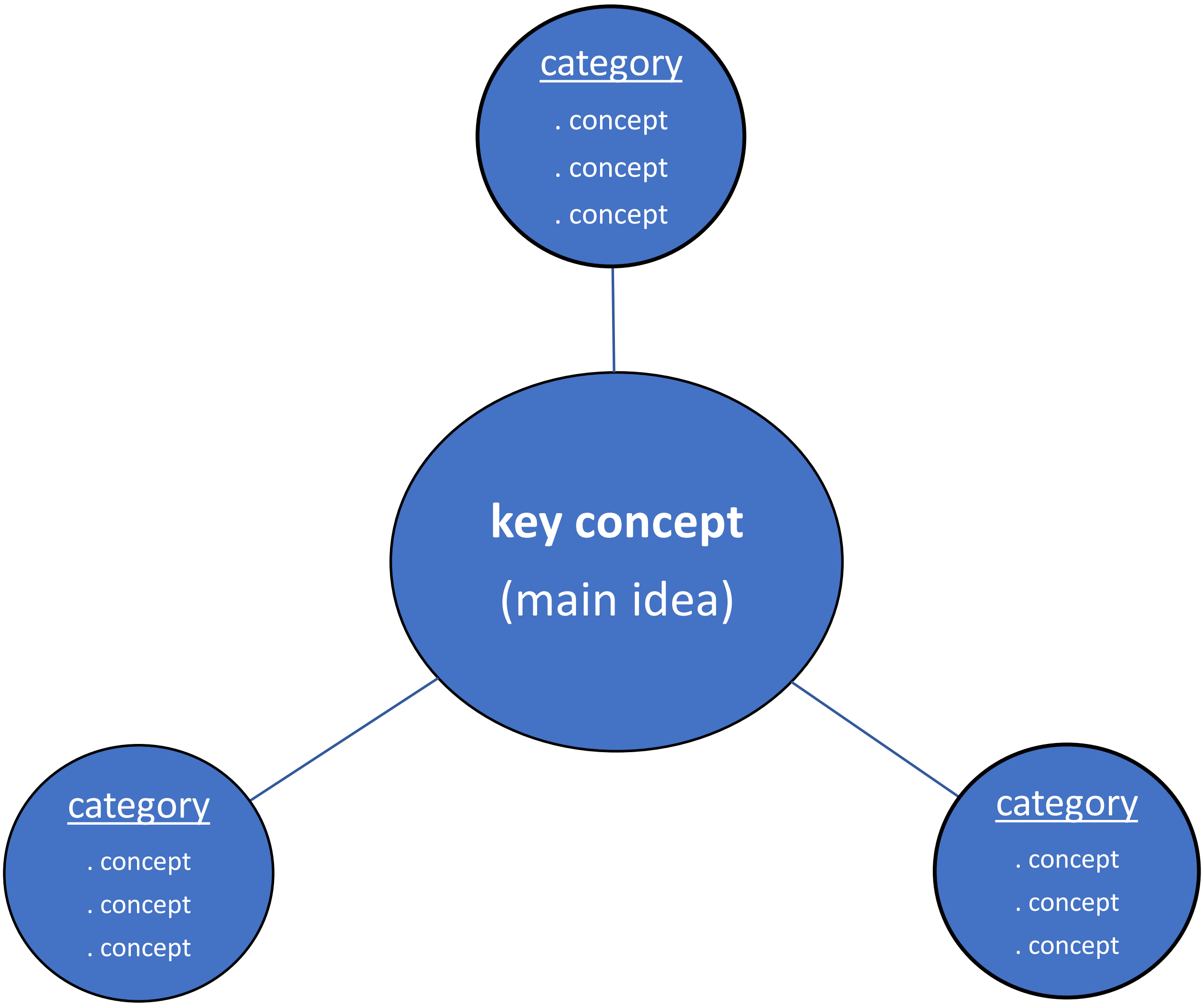

Figure.1. An example of a Semantic Map

Semantic Mapping is not a new process. It's been around for quite some time and has also been known as concept mapping, webbing, networking, and plot maps. There are various definitions of semantic mapping strategies. Antonacci (1991) describes Semantic mapping as "a visual representation of knowledge". Sinatra, Stahl, and Berg (1984) define it as "a graphic arrangement showing the major ideas and relationships in text or among word meanings" According to Stoller (1994) semantic map is "the graphic display of information within categories related to central concepts and stimulating meaningful word associations".

Semantic mapping requires the teacher and the learners to work collaboratively to construct a graphical representation map that shows the relationships between vocabulary suggested by the teacher, vocabulary suggested by the students, and vocabulary discovered in a reading text or can be used autonomously.

The use of semantic maps enables students with a deeper comprehension of words including their concept knowledge, relationships to other words, and various meanings. The mapping process stimulates students' prior knowledge of the subject and provides an effective technique to reinforce keywords, enabling students to incorporate new vocabulary into existing schemata. Semantic mapping helps the student learn new vocabulary by using known words in a semantically related network.

b. Practical Section

Many researchers have got positive outcomes with the semantic mapping strategy. Alvermann, D. E. (1981) Investigated the use of graphic organizers (Semantic Mapping) using 114 10th graders. Results showed that benefits from the use of graphic organizers were obtained. El-Koumy, A. S. (1999) Took 187 freshmen at an Egyptian university, randomly. Teacher-initiated semantic mapping; student-mediated semantic mapping The Subjects were pre-and post-tested. While the pre-test indicated no significant differences in the groups, post-test results revealed semantic mapping group scored significantly higher than the other two groups. Vakilifard et,al (2020) study recommended that semantic mapping strategy and cooperative learning approach be considered as alternative methods and instructors prefer it because it's more effective. A second group of researchers has described the effectiveness of Semantic Mapes as efficient (e. g., Berkowitz 1986, Boyle 1993, Hudson 1991, Johnson 1987, McCagg and Dansereau 1991, Ruddell and Boyle 1989). in sum, semantic mapping strategies have been continually valued by researchers as useful instructional strategies for developing reading comprehension.

The most well-known use of semantic mapping as an instructional strategy is in vocabulary building. There are two components or phases to a semantic map: introducing the topic, brainstorming, and categorization. First, Introduce the topic by choosing a core word, question, or concept: this is a keyword or phrase that is the main focus of the map. then Encourage students to come up with as many words as they can that are linked to the chosen core idea. Second Brainstorming and list subordinate ideas that help explain or clarify the main concept by categories Zaid (1995). the students grow experience in practicing some valuable cognitive skills, particularly categorizing and exemplifying and they also learn. Students can gain further practice in classification by labeling the categories on the semantic map Heimlich and Pittelman,(1986).

2. SPACE REPETITION

This section is a review that will provide evidence of the plausibility of the Space repetition, explore its efficiency, and will highlight the need for such a study to be conducted.

'The spacing effect is the observation that people tend to remember things more effectively if they use spaced repetition practice (short study periods spread out over time) as opposed to the massed practice (i.e., "cramming") Settles. S, Meeder. B, (2016). In 1885 Ebbinghaus Forgetting did a self-experimentation where he tried to memorize verbal utterances. He come up with a mathematical formula that he published in his study known as ''Memory: A Contribution to Experimental Psychology'' which first described the Ebbinghaus forgetting curve.

Research in the area of learning and memory repeatedly reveals that reading something once and then trying to recollect it results in better long-term retention than re-reading it several times. In two experiments, students studied prose passages and took one or three immediate free-recall tests another group reread the material without taking the test one week later. The group who had to recall the material without rereading it many times had greater results in the final test. Roediger & Karpicke, (2006) the study was redone in 2011 by Roediger & Butler and revealed the same results. Long-term retention improves considerably more when the delay time between each occurrence of recall gradually lengthens, as in spaced-repetition testing. When Adopting spaced repetition, a student who correctly recalls information studied a day ago would not be tested again for two days. If the student accurately remembers it at that time, the following examination will be four days later. As long as the student answers right, the time between exams increases. If the learner responds wrong, the information is retested that day until the student answers right, at which point the testing interval gradually increases again.

Flashcards are a great method for memory testing, but many language learners utilize them poorly. Although the spacing effect has been established and been proven in hundreds of experiments, It's not widely known among educators and students. Nate Kornell (2009) investigated the spacing effect in the realistic context of flashcard use. he found that Spacing was more effective than cramming—that is, massing study on the last day before the test. Across experiments, spacing was more effective than massing for 90% of the participants, yet after the first study session, 72% of the participants believed that massing had been more effective than spacing.

Many language instructors do not emphasize vocabulary in the classroom, believing that students can learn vocabulary on their own. Zimmerman, (1997). While this is true, vocabulary is more challenging and disappointing for students than grammar. Leki & Carson, (1994). The efficiency of space repetition has been proven many times in various experimental research. yet, The experimental demonstrations have not been implemented in the classroom. One significant barrier is the already full curriculum. However, the space repetition would require learners to do the spacing as assignments. Mobile devices can help in the convenience of spacing and using them outside of the classroom. The sophisticated spacing algorithms can be implemented with minimal effort on the learner's part if using a mobile device or computer.

a. Anki Application

We can and should utilize technology to structure, practice, and assess in order to optimize the rate of vocabulary acquisition. Biemiller, Andrew (2013). Chien (2015) says in a study he conducted, that users self-reported positive perceptions utilizing online flashcard apps. He states that users were motivated by the applications because it is so effective. Many commercial language learning tools, like DuoLingo, Memrise, and FluentU include spaced-repetition systems. However, this feature is only available for paid users.

The ideal tool would be a platform that is dedicated to spaced-repetition testing and is also broadly available to instructors and students. Anki is one example of this. It's an open-source and non-commercial application available for the desktop, phones, and as a website.

Anki controls the user's flashcards through spaced repetition. Since Anki manages when learners view a card for the next time, they cannot unknowingly impede their own progress by manipulating or tricking themselves as a phenomenon known as ''availability heuristic bias''

Tversky & Kahneman (1973). Anki is solely dedicated to spaced-repetition flashcards, which improves experimental validity because the effects obtained can only be attributable to this specific approach. Some language teachers, such as Bailey and Davey (2011), have included Anki into their lessons and discovered that students love using it.

III. METHOD AND DATA

A descriptive study was undertaken as a preliminary step before the main study. Quantitative data for this study was collected through a questionnaire posted student's WhatsApp group of the Department of English Studies of Ibn Zohr University. The questionnaires were aimed to identify the relationship between students' preferences for vocabulary learning strategies and the score of the vocabulary pre-test so to evaluate the progress of the subjects after combining the spaced repetition and semantic mapping.

Data were provided by 86 respondents from students. The first part was an open-ended question as to the principal means of gathering research data. This type of question permits the researcher to identify any answers the respondents provided. It was posed as Google Format to fill and was open-ended to avoid yes/no answers so that the participants, as Creswell (2005:214) suggests, could "best voice their experiences unconstrained" by any perspectives of the researcher. The second part of the questionnaire was OXFORD ONLINE ENGLISH Vocabulary Test that was linked to the Google format. These types of tests provide direct evidence that each word is actually known.

After conducting the questionnaires and the vocabulary test 16 students of English studies of ibn Zohr participated in the study. They have been gathered in two separate classrooms each containing 8 students. The first group was chosen as an Experimental Group to receive vocabulary with Semantic Mapping and used Anki later as memorizing technique, whereas the second group, the Control Group, was instructed vocabulary through traditional technique but no intervening space was used only randomly revised as they preferred after the session.

Semantic Mapping Technique and Spaced Repetition as memorizing techniques were compared to Traditional Technique to see which of the two techniques is more effective in helping students' retention of vocabulary. Figure 2 showed the Semantic map that has been used in the class. The topic was linguistics.

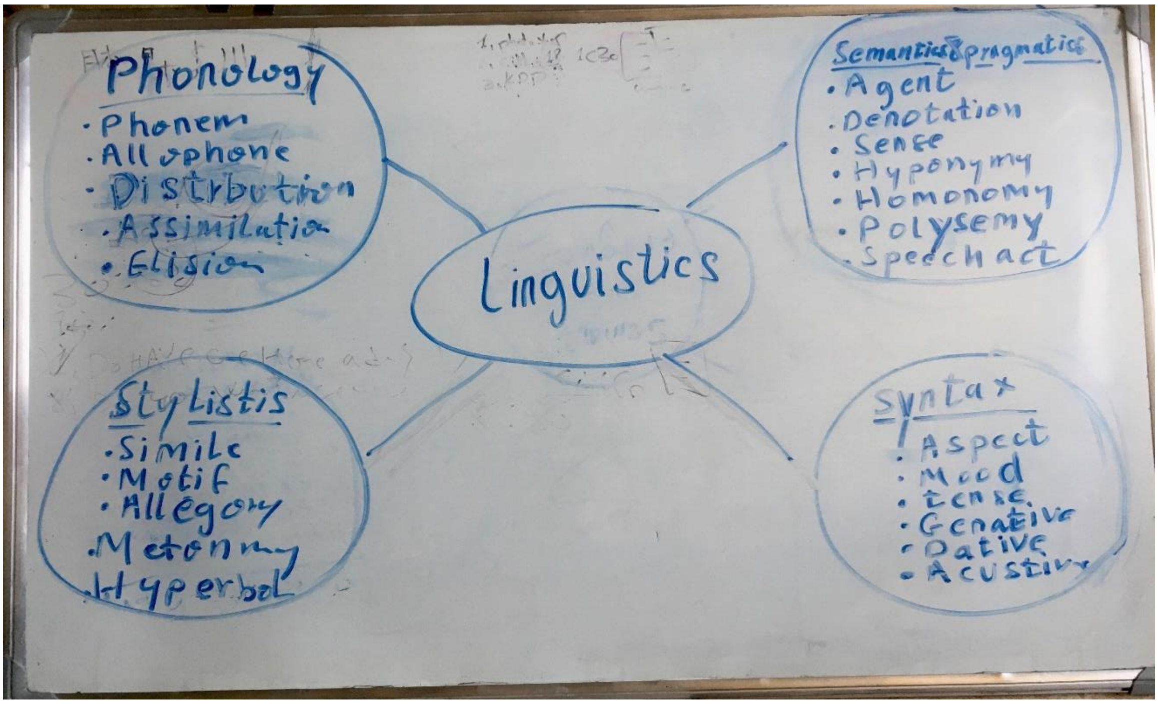

Figure.2. Semantic map

Vocabulary

Semantic & pragmatics:	Agent, denotation, Sense, hyponymy, homonymy, polysemy, speech act.
Phonology:	Phoneme, Allophone, Distribution, Free Variation, Assimilation, Elision, Epenthesis, Metathesis, Neutralization.
Syntax:	Aspect. Mood, Tens, Genitive, Dative, Accusative.
Stylistic	: Simile, Foreshadowing, Motif, Allegory, Juxtaposition, Paradox, Metonymy, Hyperbole.

This map contained the topic, suggested category labels for the target words, the target words, and suggestions for other possible categories. A post-test of the study was used after a three-week period to assess the retention rate of the target words. The test was a list of the target vocabulary items to write in the blanks in a sentence.

The validity of any test depends strongly on how seriously learners sit for the test. If they simply skip through it while playing with their cell phones, the results will be meaningless. For some learners, it may be necessary to administer the test on a one-to-one basis. Discipline and an incentive were required to complete the process of the study which most participants did not have. Students whose phones had an Android operating system downloaded the Anki Mobile application for free from the Google Play Store while the iPhone version needed payment so some students couldn't participate in the Space repetition phase.

IV. FINDINGS

The responses that participants gave in the first part of the questionnaire were converted into a pie chart to ease the analysis. The answers were grouped under four main categories according to the answers of the students: Note-taking, re-reading, using a dictionary, spaced repetition, and visualization. It seems that note-taking falls into the major vocabulary learning strategy preference and Re-reading is the second according to the date.

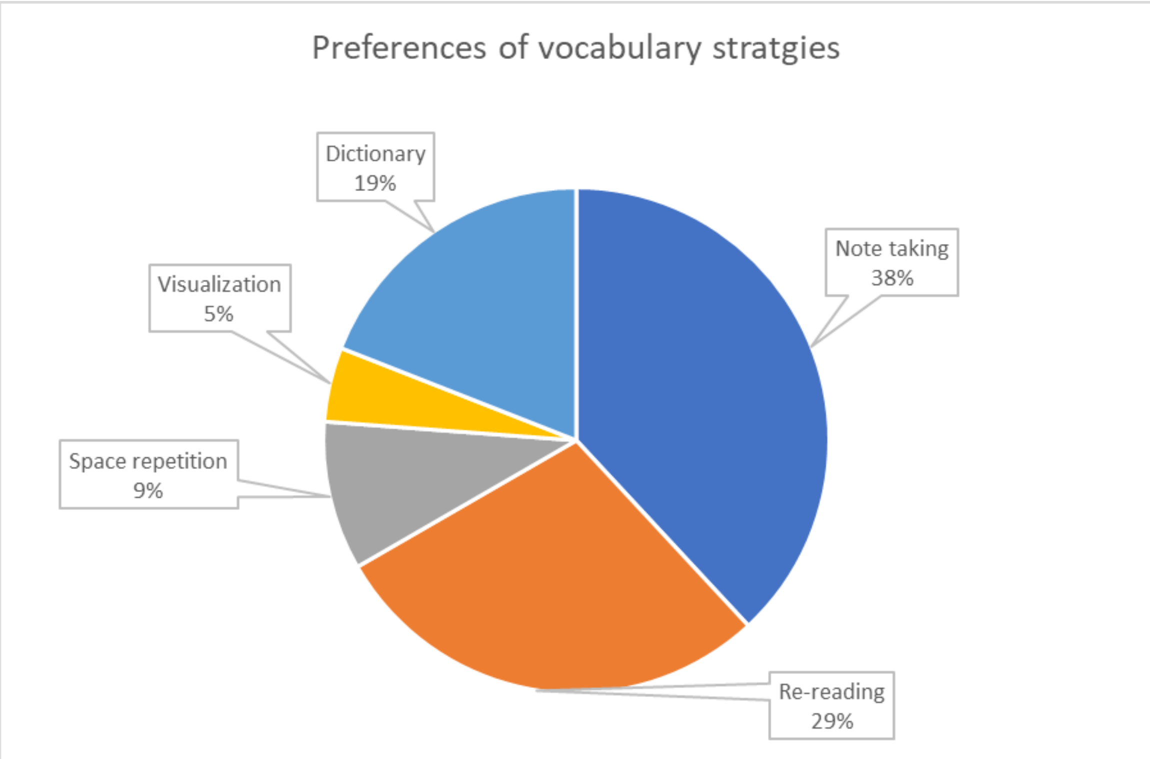

Figure.3. A pie chart of students' preference of vocabulary strategies.

Regarding the results obtained from the second part of the questionnaire, descriptive statistics are shown in figure.4. Results are reported only from 67 students who completed the process of both the questionnaire about preferred vocabulary strategy and the vocabulary test. Students who use spaced repetition as a strategy to learn new words were higher than all other categories while those who used the visualization ranked the lowest. Dictionary users scored the average. Re-reading scored 45% lower than note-taking. This finding indicates and supports that using spaced repetition in vocabulary learning improves students' achievement more than the use of other strategies.

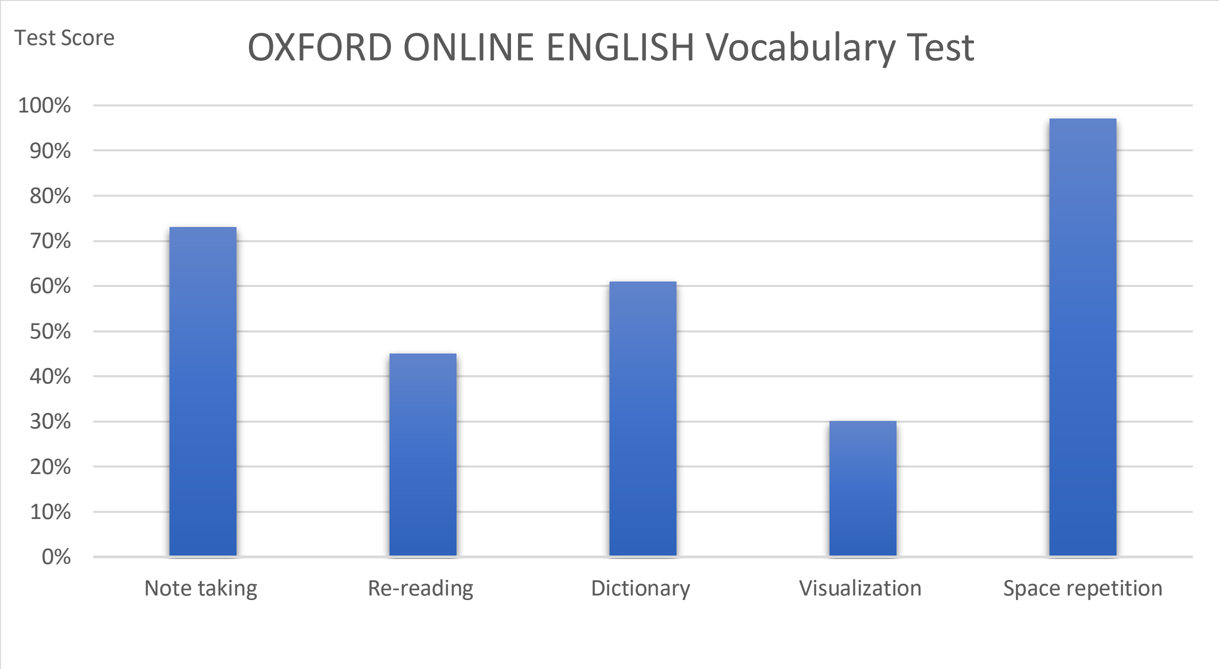

Figure.4. a bar chart represents the score of the Oxford online English vocabulary test in relation to preferred strategy.

The result post-test has shown that score of the experimental group is statistically higher then the score of the experimental group. Students in the experimental group used the vocabulary learning program Anki in their mobile phones or computer. This finding shows that the use of a vocabulary learning program improves students' vocabulary learning. Figure.5 shows the scores in percentage.

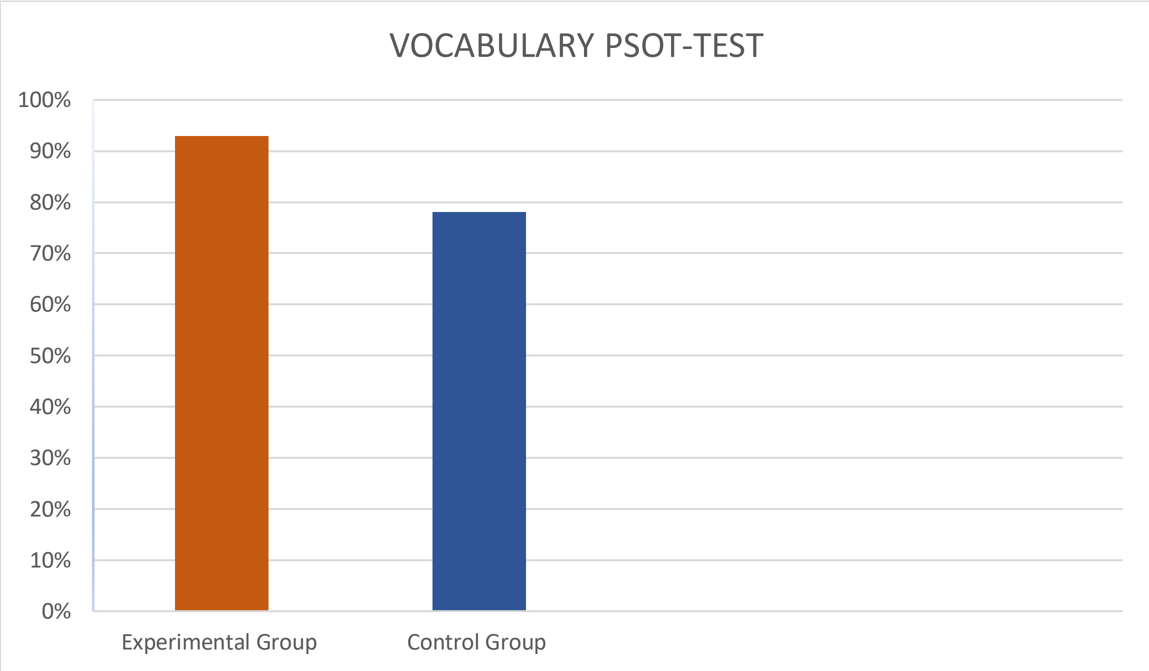

Figure.5. a bar chart of students' scores of the post-test.

V. CONCLUSION

The primary goal of learning any language is to successfully communicate using the target language. The success of the communication process cannot be assessed by the correctness of the grammatical rules; rather, the essential goal is being able to make the listener comprehend what the speaker means in the way that the speaker intends and react without disrupting the smoothness of interaction.

Of course, grammar is essential for communication efficiency; yet, the first and most crucial factor is vocabulary competency, since a speaker must know the relevant word(s). If language teachers wish to help their students enhance their vocabulary, they must be selective about the approach or strategy they choose. Many more successful approaches and strategies can be applied rather than Semantic mapping and spaced repetition. We investigated semantic mapping and space repetition as an effective strategy for developing vocabulary competency in the context of foreign language learning.

Different learners use different methods at different times and in different circumstances. To carry out a certain method cannot give proper results as all learners have different comprehension tendencies. In vocabulary learning, it is important for the learner to make effort and give the required interest. If the task does not involve enough effort, the learner may not be interested and give his/her full attention and enthusiasm

Therefore, instead of traditional techniques which are not challenging, more effective and enjoyable techniques can be more attractive for the learners. Semantic Mapping strategy combined with spaced repetition was used as an alternative way to teach vocabulary items in this study. Semantic maps were believed to assist the learner in recalling information and relating new information to prior knowledge adding the spaced repetition can max out the effectiveness of the two techniques.

In conclusion, the positive results of this study also suggest that foreign language vocabulary can be learned through semantic mapping and Spaced repetition. Semantic Mapping and Space repetition seems to be effective tool to balnce the shortness of memory capacity. It is impossible to say that one technique is completely adequate or inadequate in teaching and learning vocabulary items in foreign language and also it may be wrong to use merely one technique at all stages.

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